

CM0832 - MT5001 Elements of Set Theory  
2. Relations, Functions and Orderings

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# Outline

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Introduction

Ordered Pairs

Relations

Functions

Equivalence and Partitions

References

# Outline

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Introduction

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# Preliminaries

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## Textbook

Karel Hrbacek and Thomas Jech ([1978] 1999). Introduction to Set Theory.

## Convention

The numbers and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.

# Introduction

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## Starting...

*“We begin our program of developing set theory as a **foundation for mathematics** by showing how various general mathematical concepts, such as relations, functions, and orderings **can be represented by sets.**” (p. 17)*

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# Ordered Pairs

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## Definition [2.]1.1

Let  $a$  and  $b$  be sets. We define the **ordered pair** of  $a$  and  $b$ , denoted  $(a, b)$ , where  $a$  is the **first coordinate** and  $b$  is the **second coordinate**, using Kuratowski (1921)'s definition, that is,

$$(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}.$$

# Ordered Pairs

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$$(a, b) \stackrel{\text{def}}{=} \{\{a\}, \{a, b\}\}.$$

## Question

Is  $(a, b)$  a set?

# Ordered Pairs

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## Example

We show that  $(\emptyset, \{\emptyset\}) \neq (\{\emptyset\}, \emptyset)$ .

$$(\emptyset, \{\emptyset\}) \stackrel{\text{def}}{=} \{ \{\emptyset\}, \{\emptyset, \{\emptyset\}\} \} \neq \{ \{\{\emptyset\}\}, \{\{\emptyset\}, \emptyset\} \} \stackrel{\text{def}}{=} (\{\emptyset\}, \emptyset).$$

# Ordered Pairs

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## Example

Let  $a$  be a set. Then

$$\begin{aligned}(a, a) &\stackrel{\text{def}}{=} \{\{a\}, \{a, a\}\} \\ &\doteq \{\{a\}, \{a\}\} \\ &\doteq \{\{a\}\}.\end{aligned}$$

# Ordered Pairs

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## Exercise [2.]1.1

Let  $a$  and  $b$  sets.

- (i) Prove that  $(a, b) \in \mathcal{P}(\mathcal{P}(\{a, b\}))$ .
- (ii) Prove that  $a, b \in \bigcup(a, b)$ .
- (iii) Prove that if  $a \in A$  and  $b \in A$ , then  $(a, b) \in \mathcal{P}(\mathcal{P}(A))$ .

# Ordered Pairs

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## Theorem [2.]1.2

Let  $a, a', b, b'$  be sets. Then,

$$(a, b) \doteq (a', b') \quad \text{if and only if} \quad a \doteq a' \text{ and } b \doteq b'.$$

# Ordered Pairs

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## Theorem [2.]1.2

Let  $a, a', b, b'$  be sets. Then,

$$(a, b) \doteq (a', b') \quad \text{if and only if} \quad a \doteq a' \text{ and } b \doteq b'.$$

## Exercise

To give a different definition of ordered pair. Note that your definition must be a set and it must satisfy Theorem [2.]1.2.

# Ordered Pairs

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## Definitions

Let  $a$ ,  $b$ ,  $c$  and  $d$  sets. Then

$$(a, b, c) \stackrel{\text{def}}{=} ((a, b), c)$$

**(ordered triples)**

$$(a, b, c, d) \stackrel{\text{def}}{=} ((a, b, c), d)$$

**(ordered quadruples)**

$$(a) \stackrel{\text{def}}{=} a$$

**(one-tuples)**

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# Relations

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## Definition [2.]2.1

A set  $R$  is a **binary relation** if all elements of  $R$  are ordered pairs.

# Relations

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## Remark

*“A binary relation is, therefore, determined by specifying all ordered pairs of objects in that relation; **it does not matter by what property** the set of these ordered pairs is described.” (p. 19)*

# Relations

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## Remark

*“A binary relation is, therefore, determined by specifying all ordered pairs of objects in that relation; **it does not matter by what property** the set of these ordered pairs is described.” (p. 19)*

## Notation

Let  $R$  be a binary relation. We write  $(a, b) \in R$  or  $a R b$ .

# Relations

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## Example (informal)

Whiteboard.

# Relations

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## Example (informal)

Whiteboard.

## Example (informal)

Let  $\mathbf{N} = \{0, 1, 2, \dots\}$ . The identity relation on  $\omega$  is defined by

$$\begin{aligned}\text{Id}_{\mathbf{N}} &= \{ (n, n) \mid n \in \mathbf{N} \} \\ &= \{(0, 0), (1, 1), (2, 2), \dots\}.\end{aligned}$$

# Relations

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## Definition [2.]2.3

Let  $R$  be a binary relation. Then

$$\text{dom } R \stackrel{\text{def}}{=} \{ x \mid \text{there exists } y \text{ such as } x R y \} \quad (\text{domain of } R)$$

$$\text{ran } R \stackrel{\text{def}}{=} \{ y \mid \text{there exists } x \text{ such as } x R y \} \quad (\text{range of } R)$$

$$\text{field } R \stackrel{\text{def}}{=} \text{dom } R \cup \text{ran } R \quad (\text{field of } R)$$

# Relations

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$$\text{ran } R \stackrel{\text{def}}{=} \{ y \mid \text{there exists } x \text{ such as } x R y \} \quad (\text{range of } R)$$

$$\text{field } R \stackrel{\text{def}}{=} \text{dom } R \cup \text{ran } R \quad (\text{field of } R)$$

If  $\text{field } R \subseteq X$  then  $R$  is a relation in  $X$ .

# Relations

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## Question

Let  $R$  be a binary relation. Are  $\text{dom } R$  and  $\text{ran } R$  sets?

# Relations

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## Question

Let  $R$  be a binary relation. Are  $\text{dom } R$  and  $\text{ran } R$  sets?

## Answer

Definition of  $\text{dom } R$  and  $\text{ran } R$  without using the textbook conventions.

$$\begin{aligned}\text{dom } R &\stackrel{\text{def}}{=} \left\{ x \in \bigcup \bigcup R \mid \text{there exists } y \text{ such as } x R y \right\}, \\ \text{ran } R &\stackrel{\text{def}}{=} \left\{ y \in \bigcup \bigcup R \mid \text{there exists } x \text{ such as } x R y \right\}.\end{aligned}$$

That is,  $\text{dom } R$  and  $\text{ran } R$  are sets defined from the Axiom Scheme of Comprehension.

# Relations

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## Convention (p. 21)

When defining sets of ordered pairs we shall use the following convention:

$$\{ (x, y) \mid \mathbf{P}(x, y) \} \stackrel{\text{def}}{=} \{ w \mid w \doteq (x, y) \text{ for some } x \text{ and } y \text{ such that } \mathbf{P}(x, y) \}.$$

# Relations

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## Definition [2.]2.11

Let  $A$  be a set.

- The **membership relation** on  $A$ , denoted  $\in_A$ , is defined by

$$\in_A \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A, b \in A \text{ and } a \in b \}.$$

- The **identity relation** on  $A$ , denoted  $\text{Id}_A$ , is defined by

$$\text{Id}_A \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A, b \in A \text{ and } a \doteq b \}.$$

# Relations

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## Definition [2.]2.12

Let  $A$  and  $B$  be sets. The **Cartesian product** of  $A$  and  $B$ , denoted  $A \times B$ , is defined by

$$A \times B \stackrel{\text{def}}{=} \{ (a, b) \mid a \in A \text{ and } b \in B \}.$$

# Relations

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## Question

Let  $A$  and  $B$  be sets. Is  $A \times B$  a set?

# Relations

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## Question

Let  $A$  and  $B$  be sets. Is  $A \times B$  a set?

## Answer

Definition of  $A \times B$  without using the textbook conventions.

$$A \times B \stackrel{\text{def}}{=} \{ w \in \mathcal{P}(\mathcal{P}(A \cup B)) \mid w \doteq (a, b) \text{ for some } a \text{ and } b \text{ such that } \mathbf{P}(a, b) \},$$

where  $\mathbf{P}(x, y)$ : “ $x \in A$  and  $y \in B$ ”.

That is,  $A \times B$  is a set defined from the Axiom Scheme of Comprehension.

# Relations

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## Notation

$$A^2 \stackrel{\text{def}}{=} A \times A.$$

# Relations

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## Notation

$$A^2 \stackrel{\text{def}}{=} A \times A.$$

## Remark

Note that a binary relation  $R$  is in  $A$  if and only if  $R \subseteq A^2$ .

# Relations

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## Definition (p. 22)

Let  $A$ ,  $B$  and  $C$  be sets. The **Cartesian product** of  $A$ ,  $B$  and  $C$ , denoted  $A \times B \times C$ , is defined by

$$\begin{aligned} A \times B \times C &\stackrel{\text{def}}{=} (A \times B) \times C \\ &\doteq \{ (a, b, c) \mid a \in A, b \in B \text{ and } c \in C \}. \end{aligned}$$

# Relations

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## Definition (p. 22)

Let  $A$ ,  $B$  and  $C$  be sets. The **Cartesian product** of  $A$ ,  $B$  and  $C$ , denoted  $A \times B \times C$ , is defined by

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## Notation

$$A^3 \stackrel{\text{def}}{=} A \times A \times A.$$

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## Definition [2.]2.13

A set  $R$  is a **ternary relation** if all elements of  $R$  are ordered triples.

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## Definition [2.]2.13

A set  $R$  is a **ternary relation** if all elements of  $R$  are ordered triples.

A set  $R$  is a **ternary relation in**  $A$  if and only if  $R \subseteq A^3$ .

# Relations

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## Definition [2.]2.13

A set  $R$  is a **ternary relation** if all elements of  $R$  are ordered triples.<sup>†</sup>

A set  $R$  is a **ternary relation in**  $A$  if and only if  $R \subseteq A^3$ .

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<sup>†</sup>See the corrections to the textbook.

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## Definition (p. 22)

Let  $A$  be a set. A **unary relation in  $A$**  is any subset of  $A$ .

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# Functions

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## Definition [2.]3.1

A binary relation  $F$  is a **function** (**mapping** or **correspondence**) if

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## Definition [2.]3.1

A binary relation  $F$  is a **function** (**mapping** or **correspondence**) if

(i) (a definition without using the domain of the binary relation)

$a F b_1$  and  $a F b_2$  imply  $b_1 \doteq b_2$  for any  $a$ ,  $b_1$  and  $b_2$ .

# Functions

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for each  $a$  in  $\text{dom } F$  there is exactly one  $b$  such that  $a F b$ .

# Functions

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(ii) (a definition using the domain of the binary relation)

for each  $a$  in  $\text{dom } F$  there is exactly one  $b$  such that  $a F b$ .

## Example

Whiteboard.

# Functions

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## Notation (p. 24)

$F : A \rightarrow B \stackrel{\text{def}}{=} F$  is a function with  $\text{dom } F \doteq A$  and  $\text{ran } F \subseteq B$ .

# Functions

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## Notation (p. 24)

$F : A \rightarrow B \stackrel{\text{def}}{=} F$  is a function with  $\text{dom } F \doteq A$  and  $\text{ran } F \subseteq B$ .

## Notation (p. 24)

If  $F$  is a function with  $\text{dom } F$  we shall also use the following notations for the function  $F$ :

- (i)  $\langle F(a) \mid a \in A \rangle$ ,
- (ii)  $\langle F_a \mid a \in A \rangle$  or
- (iii)  $\langle F_a \rangle_{a \in A}$ .

# Functions

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## Discussion

Quote from the textbook:

*“Function, as understood in mathematics, is a procedure, a rule, assigning to any object  $a$  from the domain of the function a unique object  $b$ , the value of the function at  $a$ .” (p. 28)*

Is there any (implicit or explicit) procedure, rule or formula in our definition of function?

# Functions

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## Remark

The current definition of a (one-value) function (of a real variable) is from Dirichlet who in 1837 wrote:

*“If a variable  $y$  is so related to a variable  $x$  that whenever a numerical value is assigned to  $x$ , there is a **rule** according to which a unique value of  $y$  is determined, then  $y$  is said to be a **function** of the independent variable  $x$ .”*  
(Merzcbach and Boyer [1968] 2011, p. 452).

# Functions

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## Remark

The current definition of a function on arbitrary sets is from Cantor 1895:

*“By a **covering of the aggregate  $N$  with elements of the aggregate,  $M$**  or, more simply, by a **covering of  $N$  with  $M$** , we understand a **law** by which with every element  $n$  of  $N$  a definite element of  $M$  is bound up, where one and the same element of  $M$  can come repeatedly into application. The element of  $M$  bound up with  $n$  is, in a way, a one-valued function of  $n$ , and may be denoted by  $f(n)$ ; it is called a **covering function of  $n$** . The corresponding covering of  $N$  will be called  $f(N)$ .” (Cantor [1895] 1915, p. 94)*

# Functions

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## Lemma [2.]3.2 (function extensionality)

Let  $F$  and  $G$  be functions.  $F \doteq G$  if and only if  $\text{dom } F \doteq \text{dom } G$  and  $F(x) \doteq G(x)$ , for all  $x \in \text{dom } F$ .

# Functions

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## Definitions [2.]3.3

Let  $F$  be a function and  $A$  and  $B$  sets.

- (i)  $F$  is a function **on** (**from**)  $A$  if and only if  $\text{dom } F \doteq A$ .
- (ii)  $F$  is a function **into** (**to**)  $B$  if and only if  $\text{ran } F \subseteq B$ .
- (iii)  $F$  is a function **onto**  $B$  if and only if  $\text{ran } F \doteq B$ .
- (iv) The **restriction** of  $F$  to  $A$ , denoted  $F \upharpoonright A$ , is the function defined by

$$F \upharpoonright A \stackrel{\text{def}}{=} \{ (x, y) \in F \mid x \in A \}.$$

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## Definition [2.]3.7

A function  $F$  is **one-to-one** (or **injective**) if for each  $y \in \text{ran } F$  there is only one  $x$  such that  $x F y$ . In other words, if  $x_1, x_2 \in \text{dom } F$  and  $x_1 \neq x_2$  implies  $f(x_1) \neq f(x_2)$ .

# Functions

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## Example

Whiteboard.

# Functions

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## Definition [2.]3.13

Let  $A$  and  $B$  be sets. The **set of functions** on  $A$  into  $B$ , denoted  $B^A$ , is defined by

$$B^A \stackrel{\text{def}}{=} \{ F \mid F : A \rightarrow B \}.$$

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## Definition [2.]3.13

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## Example

- $\{0, 1\}^{\mathbb{N}}$ : The set of infinity binary sequences.

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- $\{0, 1\}^{\mathbb{N}}$ : The set of infinity binary sequences.
- $\emptyset^A \doteq \emptyset$  for  $A \neq \emptyset$  (no function can have a non-empty domain and an empty range).

# Functions

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$$B^A \stackrel{\text{def}}{=} \{ F \mid F : A \rightarrow B \}.$$

## Example

- $\{0, 1\}^{\mathbb{N}}$ : The set of infinity binary sequences.
- $\emptyset^A \doteq \emptyset$  for  $A \neq \emptyset$  (no function can have a non-empty domain and an empty range).
- $A^\emptyset \doteq \{\emptyset\}$  for any set  $A$  ( $\emptyset$  is the only function with an empty domain).

# Functions

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## Question

Let  $A$  and  $B$  be sets. Is  $B^A$  a set?

# Functions

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## Question

Let  $A$  and  $B$  be sets. Is  $B^A$  a set?

## Answer

Definition of  $B^A$  without using the textbook conventions.

$$B^A \stackrel{\text{def}}{=} \{ F \in \mathcal{P}(A \times B) \mid F : A \rightarrow B \}.$$

That is,  $B^A$  is a set defined from the Axiom Scheme of Comprehension.

# Functions

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## Convention (p. 27)

Let  $S = \langle S_i \mid i \in I \rangle$  be a function with domain  $I$ .

*“We call the function  $\langle S_i \mid i \in I \rangle$  an **indexed system of sets**, whenever we wish to stress that the values of  $S$  are sets.”*

# Functions

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## Definition (p. 27)

Let  $S = \langle S_i \mid i \in I \rangle$  be an indexed system of sets. The **product** (or **generalised product**) of  $S$ , denoted  $\prod S$ , is defined by

$$\begin{aligned} \prod S &\stackrel{\text{def}}{=} \{ f \mid f \text{ is a function on } I \text{ and } f_i \in S_i, \text{ for all } i \in I \} \\ &\doteq \left\{ f \mid f : I \rightarrow \bigcup \{ S_i \mid i \in I \} \text{ and } f_i \in S_i, \text{ for all } i \in I \right\}. \end{aligned}$$

# Functions

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## Example (informal)

Let  $A$  and  $B$  be two sets,  $I = \{0, 1\}$  and let  $S = \langle S_i \mid i \in I \rangle$  be an indexed system of sets where  $S_0 = A$  and  $S_1 = B$ . Then

$$\prod S \doteq \left\{ f \mid f : I \rightarrow \bigcup \{A, B\}, \text{ such that } f_0 \in A \text{ and } f_1 \in B \right\}.$$

Now, we can define the one-to-one correspondence

$$\begin{aligned} h : \prod S &\rightarrow A \times B \\ h(f) &= (f_0, f_1). \end{aligned}$$

# Functions

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## Remark

Let  $\langle S_i \mid i \in I \rangle$  be an indexed system of sets. If  $S_i = B$  for all  $i \in I$ , then

$$\begin{aligned}\prod S &\doteq B^I \\ &\doteq \{ f \mid f : I \rightarrow B \}.\end{aligned}$$

# Functions

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## Notation (p. 27)

Let  $\prod S$  be the product of an indexed system of sets  $S = \langle S_i \mid i \in I \rangle$ . We shall also use the following notations for the product  $\prod S$ :

- (i)  $\prod \langle S_i \mid i \in I \rangle$ ,
- (ii)  $\prod_{i \in I} S(i)$  or
- (iii)  $\prod_{i \in I} S_i$ .

# Functions

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## Question

Let  $\prod S$  be the product of an indexed system of sets  $S = \langle S_i \mid i \in I \rangle$ .

Is  $\prod S$  a set?

# Functions

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## Question

Let  $\prod S$  be the product of an indexed system of sets  $S = \langle S_i \mid i \in I \rangle$ .

Is  $\prod S$  a set?

## Answer

Definition of  $\prod S$  without using the textbook conventions.

$$\prod S \stackrel{\text{def}}{=} \left\{ f \in \mathcal{P} \left( I \times \bigcup \{ S_i \mid i \in I \} \right) \mid f \text{ is a function on } I \text{ and } f_i \in S_i, \text{ for all } i \in I \right\}$$

That is,  $\prod S$  is a set defined from the Axiom Scheme of Comprehension.

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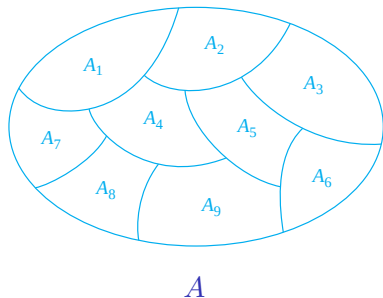
Functions

**Equivalence and Partitions**

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# Equivalence and Partitions

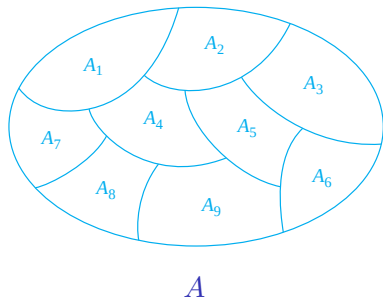
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- (i)  $A_i \cap A_j = \emptyset$  for  $i \neq j$ ,
- (ii)  $A = \bigcup A_i$ .

# Equivalence and Partitions<sup>†</sup>

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- (i)  $A_i \cap A_j = \emptyset$  for  $i \neq j$ ,
- (ii)  $A = \bigcup A_i$ .

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<sup>†</sup>Figure source: (Rosen [1988] 2012, Fig. § 9.5, Fig. 1).

# Equivalence and Partitions

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## Definition [2.]4.1

Let  $R$  be a binary relation in a set  $A$ . The relation  $R$  is

- (a) **reflexive in  $A$**  if for all  $x \in A$ ,  $x R x$ ;
- (b) **symmetric in  $A$**  if for all  $x, y \in A$ ,  $x R y$  implies  $y R x$ .
- (c) **transitive in  $A$**  if for all  $x, y, z \in A$ ,  $x R y$  and  $y R z$  imply  $x R z$ ;
- (d) an **equivalence on  $A$**  if  $R$  is reflexive, symmetric and transitive in  $A$ .

# Equivalence and Partitions

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## Questions

(i) Let  $A = \{a, e, i, o, u\}$ . Is the identity relation in  $A$  an equivalence on  $A$ ?

# Equivalence and Partitions

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## Questions

- (i) Let  $A = \{a, e, i, o, u\}$ . Is the identity relation in  $A$  an equivalence on  $A$ ?
- (ii) Let  $A \neq \emptyset$  be a set. Is the relation  $\emptyset$  an equivalence on  $A$ ? What about if  $A \doteq \emptyset$ ?

# Equivalence and Partitions

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## Questions

- (i) Let  $A = \{a, e, i, o, u\}$ . Is the identity relation in  $A$  an equivalence on  $A$ ?
- (ii) Let  $A \neq \emptyset$  be a set. Is the relation  $\emptyset$  an equivalence on  $A$ ? What about if  $A \doteq \emptyset$ ?
- (iii) Let  $A \neq \emptyset$  be a set. Is the relation  $A \times A$  an equivalence on  $A$ ? What about if  $A \doteq \emptyset$ ?

# Equivalence and Partitions

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## Questions

- (i) Let  $A = \{a, e, i, o, u\}$ . Is the identity relation in  $A$  an equivalence on  $A$ ?
- (ii) Let  $A \neq \emptyset$  be a set. Is the relation  $\emptyset$  an equivalence on  $A$ ? What about if  $A \doteq \emptyset$ ?
- (iii) Let  $A \neq \emptyset$  be a set. Is the relation  $A \times A$  an equivalence on  $A$ ? What about if  $A \doteq \emptyset$ ?
- (iv) Let  $A$  be a singleton. It is possible to define an equivalence on  $A$ ?

# Outline

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Introduction

Ordered Pairs

Relations

Functions

Equivalence and Partitions

References

# References

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